

Literature Status: Two Roe Rigidity Questions

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Status

literature_already_answered. This is a lightweight status packet, not a new proof claim by the run. Both questions on page 18 of the source paper are consequences of later general rigidity theorems.

Source Questions

The source is Bruno M. Braga, Yeong Chyuan Chung, and Kang Li, *Coarse Baum–Connes conjecture and rigidity for Roe algebras*, arXiv:1907.10237, J. Funct. Anal. 279 (2020), 108728 [1]. On page 18, Question 5.5 asks:

There is a finitely generated group Γ which contains an expander X with large girth isometrically in its Cayley graph. Thus Γ does not satisfy the coarse Baum–Connes conjecture with coefficients, and the sparse subspace X yields a noncompact ghost projection. Is Γ Roe rigid?

On the same page, Question 5.8 asks:

Let X be a uniformly locally finite metric space which admits a coarse embedding into a Hilbert space. Is X necessarily uniform Roe bijectively rigid?

Identification

For Question 5.5, Diego Martinez and Federico Vigolo prove in Theorem A of *C*-Rigidity of bounded geometry metric spaces*, arXiv:2501.03128, page 2 [2], that if uniformly locally finite metric spaces X and Y have isomorphic Roe algebras, then X and Y are coarsely equivalent. A Cayley graph of a finitely generated group is uniformly locally finite. Therefore every such Γ , including the group in Question 5.5, is Roe rigid. Martinez–Vigolo cite the source paper as reference [6] and describe their theorem as the complete solution of this C*-rigidity problem.

For Question 5.8, Alessandro Vignati proves in Theorem A of *The rigidity problem for uniform Roe algebras*, arXiv:2602.01998, page 3 [3], that if the uniform Roe algebras of uniformly locally finite metric spaces X and Y are isomorphic, then X and Y are bijectively coarsely equivalent. Thus every uniformly locally finite X is uniform Roe bijectively rigid. Question 5.8 has a positive answer, with no coarse Hilbert embedding assumption needed.

Scope and Search Evidence

The run's lightweight registry, solution, attempt, and proof-gap indexes were searched for arXiv:1907.10237 and the two rigidity phrases. No exact prior record was found. The decisive later sources are arXiv:2501.03128, whose Theorem A solves Roe rigidity generally and whose bibliography cites the source, and arXiv:2602.01998, whose abstract, introduction, and Theorem A state the unconditional solution of uniform Roe bijective rigidity. No part of Questions 5.5 or 5.8 remains open after these theorems.

References

- [1] Bruno M. Braga, Yeong Chyuan Chung, and Kang Li, *Coarse Baum–Connes conjecture and rigidity for Roe algebras*, J. Funct. Anal. 279 (2020), 108728; arXiv:1907.10237.
- [2] Diego Martinez and Federico Vigolo, *C^* -Rigidity of bounded geometry metric spaces*, Publ. Math. IHES 141 (2025), 333–348; arXiv:2501.03128.
- [3] Alessandro Vignati, *The rigidity problem for uniform Roe algebras*, arXiv:2602.01998, 2026.